

Concrete Mix Design Report – M25

Project: Residential RCC Structure **Concrete Grade:** M25

Material / Parameter	Value	Unit	Remarks
Cement	360	kg/m³	OPC/PPC equivalent cementitious binder
Fly Ash	60	kg/m³	Supplementary cementitious material
Water	170	L/m³	Batch water before moisture correction
Fine Aggregate / Sand	735	kg/m³	Zone-dependent natural/manufactured sand
10 mm Coarse Aggregate	430	kg/m³	Nominal coarse aggregate fraction
20 mm Coarse Aggregate	650	kg/m³	Nominal coarse aggregate fraction
Chemical Admixture	4.0	kg/m³	High-range water reducer
Water/Cementitious Ratio	0.405	-	Illustrative value
Target Slump	90	mm	At point of discharge
Target Mean Strength	31.6	MPa	Illustrative 28-day target

Design Notes

The mix is intended for general reinforced-concrete applications. Aggregate quantities are illustrative and should be corrected for aggregate moisture and absorption before production.

Document status: Synthetic sample created for RMC RAG development and testing. Not an approved construction mix design. Actual proportions require laboratory validation, material test results, project specifications, moisture corrections, and applicable standards.